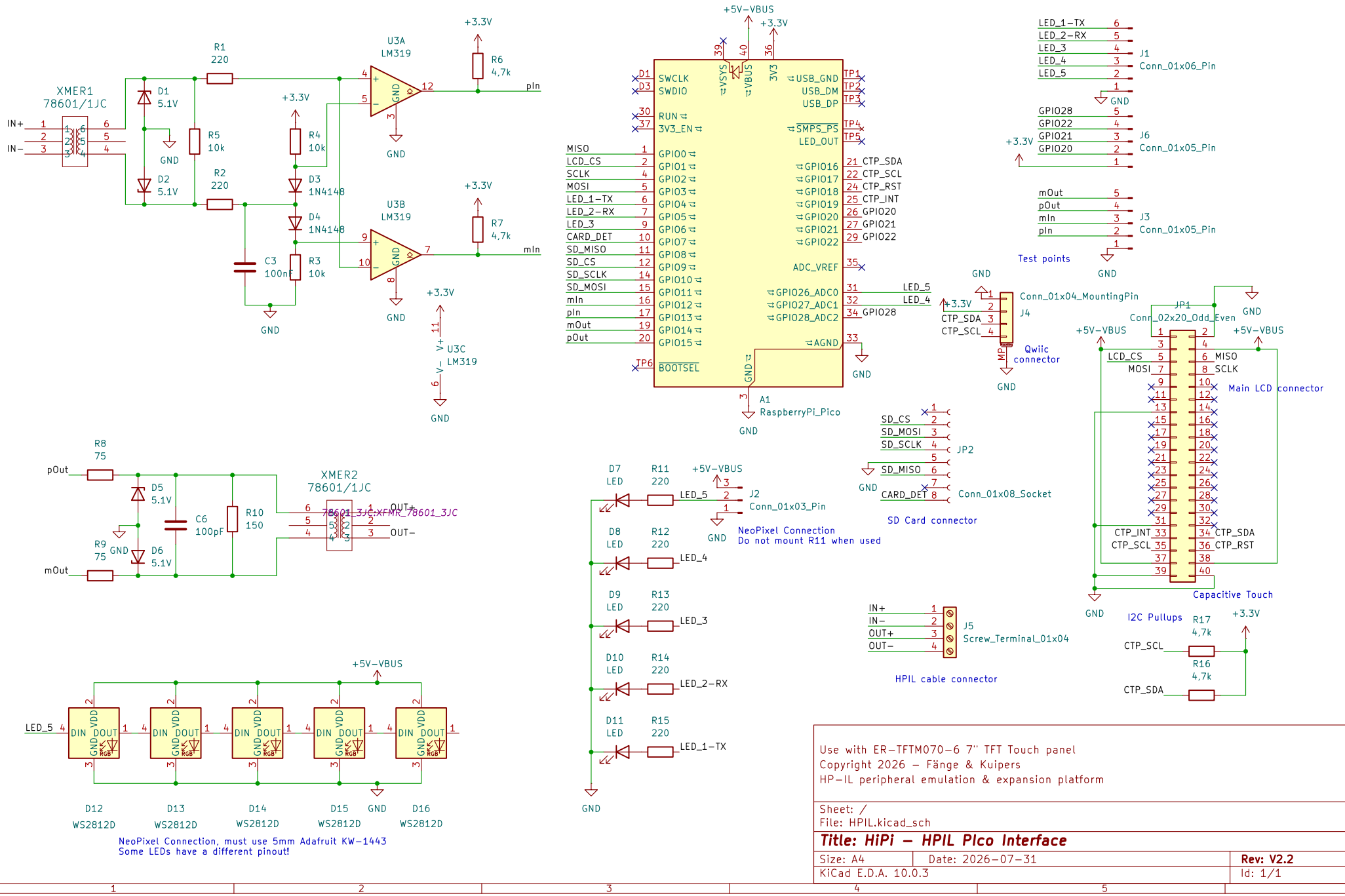

XMER1
78601/1JC
IN+
IN−
D1
5.1V
D2
5.1V
R1
220
R2
220
R5
10k
C3
100nF
GND
U3A
LM319
U3B
LM319
R4
10k
R3
10k
D3
1N4148
D4
1N4148
+3.3V
R6
4,7k
R7
4,7k
pIn
mIn
U3C
LM319
V+
V−
+5V−VBUS
A1
RaspberryPi_Pico
SWCLK
SWDIO
RUN
3V3_EN
BOOTSEL
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15
MISO
LCD_CS
SCLK
MOSI
LED_1−TX
LED_2−RX
LED_3
CARD_DET
SD_MISO
SD_CS
SD_SCLK
SD_MOSI
mIn
pIn
mOut
pOut
USB_GND
USB_DM
USB_DP
SMPS_PS
LED_OUT
ADC_VREF
GPIO16
GPIO17
GPIO18
GPIO19
GPIO20
GPIO21
GPIO22
GPIO26_ADC0
GPIO27_ADC1
GPIO28_ADC2
AGND
CTP_SDA
CTP_SCL
CTP_RST
CTP_INT
LED_5
LED_4
GPIO28
J1
Conn_01x06_Pin
J6
Conn_01x05_Pin
J3
Conn_01x05_Pin
Test points
J4
Conn_01x04_MountingPin
Qwiic
connector
JP1
Conn_02x20_Odd_Even
Main LCD connector
Capacitive Touch
I2C Pullups
R17
4,7k
R16
4,7k
JP2
Conn_01x08_Socket
SD Card connector
J5
Screw_Terminal_01x04
HPIL cable connector
J2
Conn_01x03_Pin
NeoPixel Connection
Do not mount R11 when used
R8
75
R9
75
R10
150
C6
100pF
D5
5.1V
D6
5.1V
XMER2
78601/1JC
78601_3JC:XFMR_78601_3JC
pOut
mOut
OUT+
OUT−
D7
LED
D8
LED
D9
LED
D10
LED
D11
LED
R11
220
R12
220
R13
220
R14
220
R15
220
LED_5
LED_4
LED_3
LED_2−RX
LED_1−TX
D12
WS2812D
D13
WS2812D
D14
WS2812D
D15
WS2812D
D16
WS2812D
DIN
DOUT
GND
VDD
NeoPixel Connection, must use 5mm Adafruit KW−1443
Some LEDs have a different pinout!
Use with ER−TFTM070−6 7" TFT Touch panel
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HP−IL peripheral emulation & expansion platform
Sheet: /
File: HPIL.kicad_sch
Title: HiPi − HPIL Pico Interface
Size: A4
Date: 2026−07−31
Rev: V2.2
KiCad E.D.A. 10.0.3
Id: 1/1